

Designing Secure and Resilient Network Architectures

This session will provide an overview of key concepts and best practices for designing network infrastructures that are both secure and highly available.

What is Network Architecture?



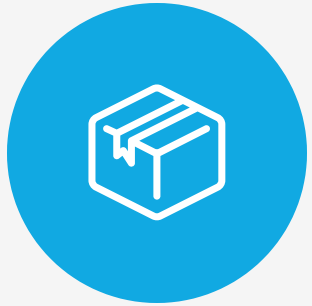
Defines the overall structure and design

Network architecture outlines the communication infrastructure and systems within an organization, determining how data is transmitted, stored, and protected.



Dictates data transmission and storage

Network architecture prescribes the protocols, technologies, and methodologies used to move and store data across the organization's communication networks.



Includes security measures

Network architecture incorporates security controls such as segmentation, redundancy, perimeter defense, and monitoring to protect against unauthorized access and cyber threats.

By establishing a well-structured network architecture, organizations can ensure optimal performance, security, and scalability of their communication infrastructure.

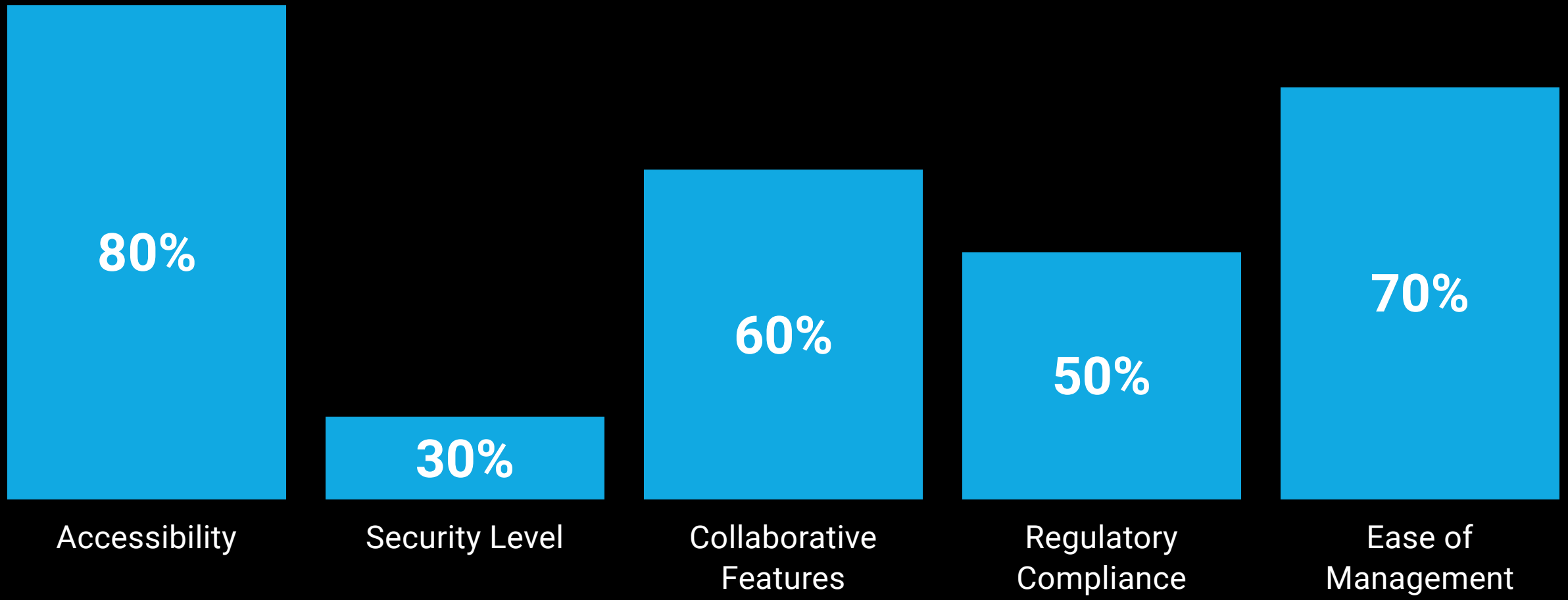
Importance of Redundancy and Availability

Prevent Single Points of Failure	Ensure Business Continuity	Achieve Network Resilience	Implement Automated Failover	Conduct Regular Testing
Redundant systems and high availability architectures ensure that there are no single points of failure in the network, allowing operations to continue even in the event of hardware failures, cyberattacks, or natural disasters.	By implementing redundancy and high availability measures, organizations can maintain uninterrupted service and minimize downtime, ensuring seamless business operations and protecting against disruptions that could impact revenue, productivity, and customer satisfaction.	Redundant network components, diverse routing paths, and geographically distributed data centers contribute to a resilient network architecture that can withstand and recover from various types of failures or attacks, safeguarding the organization's critical infrastructure and data.	Organizations should deploy automated failover solutions that can quickly and seamlessly switch to backup systems or alternate paths, ensuring that services remain available and accessible to users, even in the event of a primary system or component failure.	Regularly testing the redundancy and high availability mechanisms is crucial to ensure that they function as intended and can effectively mitigate potential disruptions, allowing organizations to identify and address any vulnerabilities or weaknesses in the network architecture.

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Distinguishing Internet and Intranet

Comparison of key characteristics (0-100 scale)



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Extranet: Secure External Collaboration

What is an Extranet?

An extranet extends an organization's private network to allow controlled access to external partners, vendors, and clients. It enables secure collaboration and information sharing beyond internal employees.

Access Control Policies

Extranets implement strict access control policies to define which external users can access specific extranet resources. Authentication protocols and role-based access controls ensure only authorized individuals can interact with sensitive data.

Network Segmentation

Extranets utilize network segmentation to isolate extranet resources from the internal network. This prevents external users from accessing or compromising internal systems, maintaining a strong security perimeter.

Encryption and Firewalls

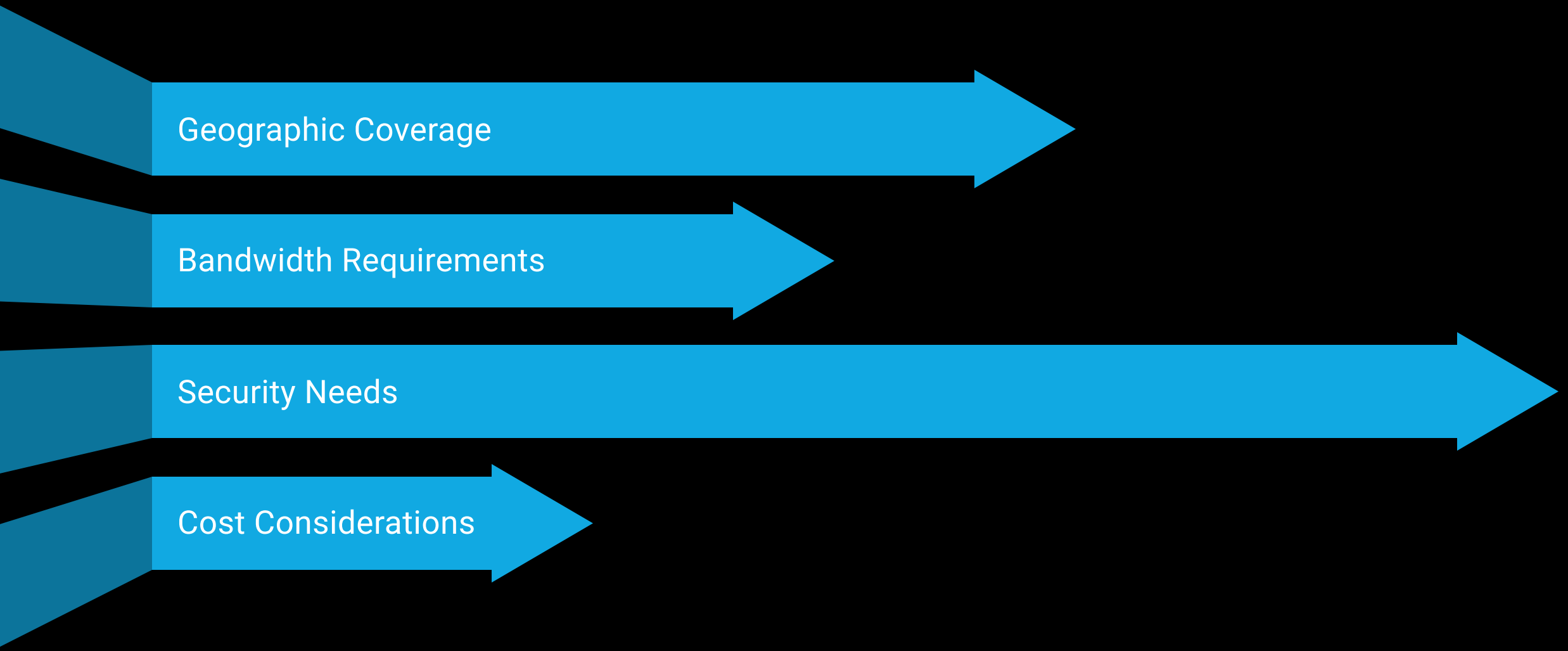
Extranet communications are encrypted to ensure confidentiality and integrity of data shared with external partners. Firewalls are deployed at the extranet boundary to monitor and filter traffic, blocking unauthorized access attempts.

Cybersecurity Measures

Organizations implement robust cybersecurity measures for their extranets, including intrusion detection, logging, and regular security audits. These controls help identify and mitigate potential threats from external collaborators.

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Choosing the Right Network Type



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Establishing Perimeter Controls

Firewalls, intrusion detection systems (IDS), and access control lists (ACLs) serve as the first line of defense in securing network boundaries and limiting unauthorized access. These perimeter controls inspect and filter network traffic based on predefined security policies, preventing malicious activity from reaching internal systems.

Securing Modems: A Potential Vulnerability

- ## Modem Vulnerabilities
  Attackers can exploit vulnerable modems to gain unauthorized access to networks, intercept communications, or launch denial-of-service attacks.
- ## Authentication Measures
  Implement strong authentication mechanisms, such as multi-factor authentication, to prevent unauthorized access to modems.
- ## Encryption Protocols
  Ensure that all modem communications are encrypted using industry-standard encryption protocols to protect data confidentiality and integrity.
- ## Firewall Protections
  Deploy firewall rules to monitor and control traffic to and from modems, blocking unauthorized access and suspicious activity.
- ## Firmware Updates
  Regularly update modem firmware to address known vulnerabilities and implement the latest security patches.
- ## Disable Unnecessary Features
  Disable any unnecessary modem features to minimize the attack surface and reduce the risk of exploitation.
- ## Monitoring and Auditing
  Continuously monitor modem activity and conduct regular security audits to detect and address any suspicious behavior or unauthorized access attempts.

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Communications and Network Policies

- **Acceptable Use Policy**
  Defines guidelines for appropriate use of network resources and communication tools, including internet and email usage.
- **Access Control Mechanisms**
  Implements authentication, authorization, and role-based access controls to ensure only authorized users can access sensitive data and systems.
- **Encryption Standards**
  Establishes requirements for data encryption, both in transit and at rest, to protect confidentiality and integrity of information.
- **Remote Access Policy**
  Defines secure procedures and protocols for remote workers to access internal resources, including the use of VPNs and multi-factor authentication.
- **Regulatory Compliance**
  Ensures network and communication policies align with relevant industry regulations and data privacy laws to avoid legal and financial penalties.

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Firewalls: The Gatekeepers of Network Security

Firewalls Defined

Firewalls are critical components of network security that control traffic flow between trusted and untrusted networks. They enforce security policies by filtering incoming and outgoing traffic based on predefined rules.

Multi-Layer Protection

Firewalls operate at different layers of the OSI model, with packet-filtering firewalls inspecting individual packets and application-layer firewalls analyzing traffic at the application level to provide comprehensive security.

Hardware vs. Software Firewalls

Organizations can deploy hardware or software firewalls, depending on network size and security requirements. Firewall configurations must be regularly updated to address emerging threats and prevent unauthorized access.

Firewalls and Routers

Firewalls and routers serve different roles in network security. Firewalls enforce security policies to control traffic flow, while routers facilitate the routing of data packets between networks. In secure network architectures, firewalls are deployed alongside routers to provide layered security.

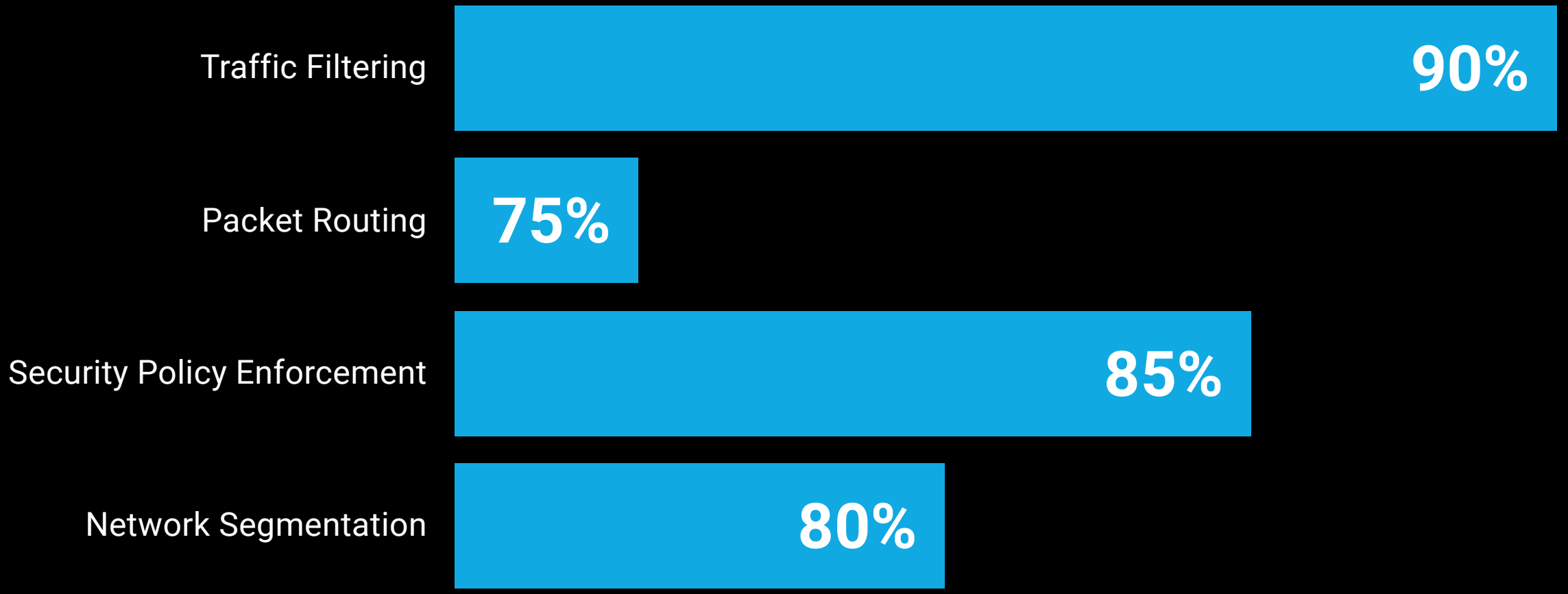
DMZ Perimeter Controls

A demilitarized zone (DMZ) is a segregated network area that hosts public-facing services. DMZ perimeter controls, such as firewalls, intrusion prevention systems, and network segmentation, restrict direct access between external users and internal systems, reducing the risk of unauthorized access.

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Firewalls vs. Routers: Complementary Roles

Comparing the primary functions of firewalls and routers



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Demilitarized Zone (DMZ) Perimeter Controls

- ## Segregated Network Area
  The DMZ is a dedicated network segment that hosts public-facing services, isolating them from the internal network.

- ## Public-facing Services
  The DMZ hosts web servers, email gateways, DNS servers, and other services that need to be accessible from the internet.

- ## Firewall Protection
  Firewalls are deployed at the DMZ's perimeter to control and filter incoming and outgoing traffic, enforcing access policies.

- ## Intrusion Prevention System (IPS)
  IPS solutions monitor network traffic and actively block suspicious activity in real-time, enhancing the DMZ's security.

- ## Network Segmentation
  The DMZ is logically and physically separated from the internal network, limiting the potential impact of attacks.

Intrusion Detection and Prevention Systems (IDS/IPS)

- **Intrusion Detection Systems (IDS)**
  Passively analyze network traffic and generate alerts when potential threats are detected
- **Intrusion Prevention Systems (IPS)**
  Actively block suspicious activity in real-time to prevent attacks and enhance network security
- **IDS/IPS Integration**
  IDS and IPS solutions should integrate with firewalls and SIEM platforms to improve threat visibility and response capabilities
- **Comprehensive Monitoring**
  Combining host-based IDS (HIDS) and network-based IDS (NIDS) provides comprehensive network security monitoring
- **Automated Threat Response**
  IPS can automatically take corrective actions such as blocking IP addresses, terminating sessions, or modifying firewall rules

IDS Architecture

Host-based IDS (HIDS)

HIDS monitors activity on individual devices, such as servers and endpoints, detecting suspicious behavior at the operating system level. It provides visibility into host-specific events and can identify malware infections or unauthorized access attempts on individual systems.

Network-based IDS (NIDS)

NIDS analyzes network traffic, identifying anomalies and signature-based threats. It examines the flow of data across the network to detect potential intrusion attempts, unauthorized access, or the presence of malicious activity.

Hybrid IDS Approach

A hybrid IDS combines both host-based and network-based detection mechanisms, providing a comprehensive network security monitoring solution. This approach leverages the strengths of both HIDS and NIDS to offer a more robust and effective defense against a wide range of cyber threats.

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Intrusion Prevention System (IPS)

An intrusion prevention system (IPS) is a proactive security solution that analyzes network traffic in real-time and takes immediate corrective actions to block malicious activity. Unlike intrusion detection systems (IDS) that only generate alerts, an IPS can automatically intervene to protect the network by identifying and mitigating threats before they can cause harm.



Security Information & Event Management (SIEM)

Centralized Log Aggregation

SIEM solutions collect and aggregate security logs from various network devices, including firewalls, IDS/IPS, and endpoints.

Threat Detection and Analysis

SIEM platforms correlate and analyze the aggregated logs to identify potential security threats, such as unauthorized access attempts, malware infections, and anomalous activities.

Real-time Threat Intelligence

SIEM provides real-time visibility into the security posture, enabling organizations to quickly detect and respond to emerging threats.

Efficient Incident Response

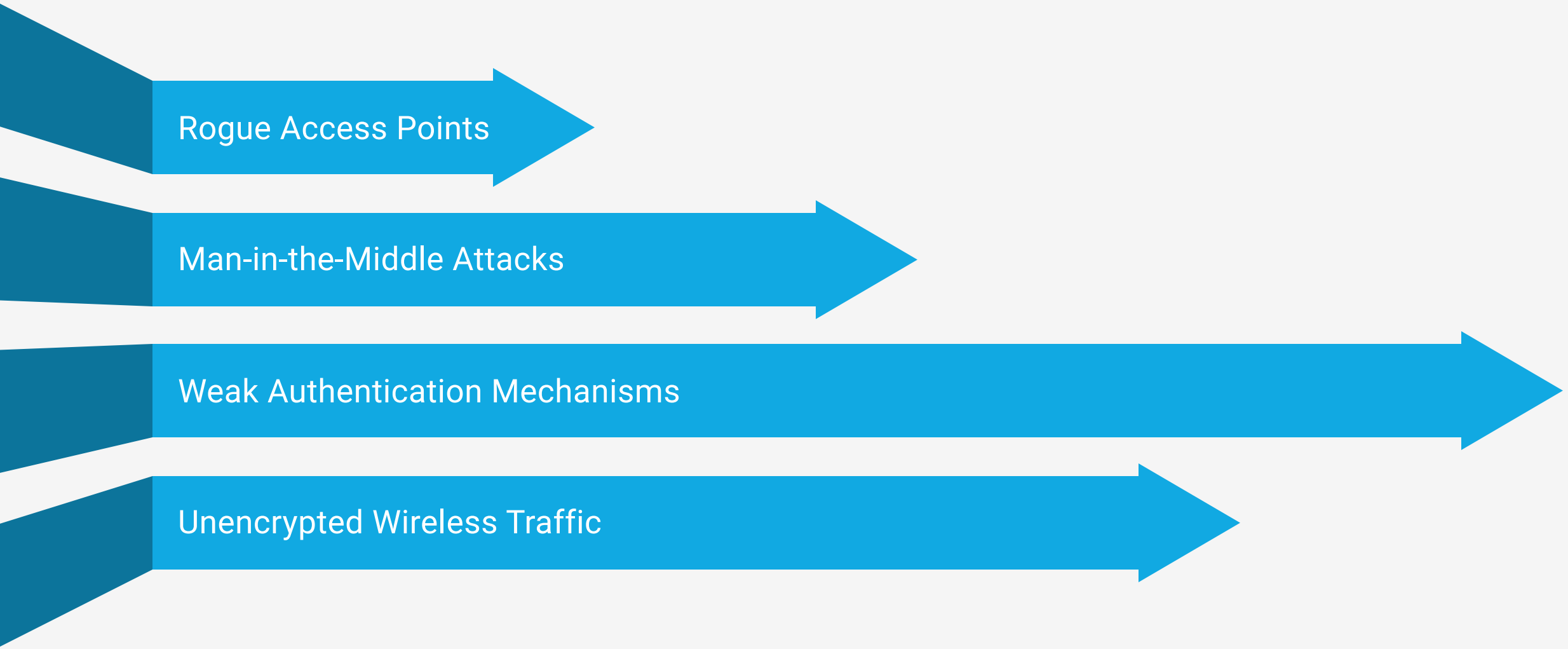
By integrating with security tools and automating response workflows, SIEM solutions help organizations streamline the incident response process and minimize the impact of security breaches.

Compliance Reporting

SIEM platforms generate comprehensive reports for compliance and regulatory requirements, such as PCI DSS, HIPAA, and GDPR.

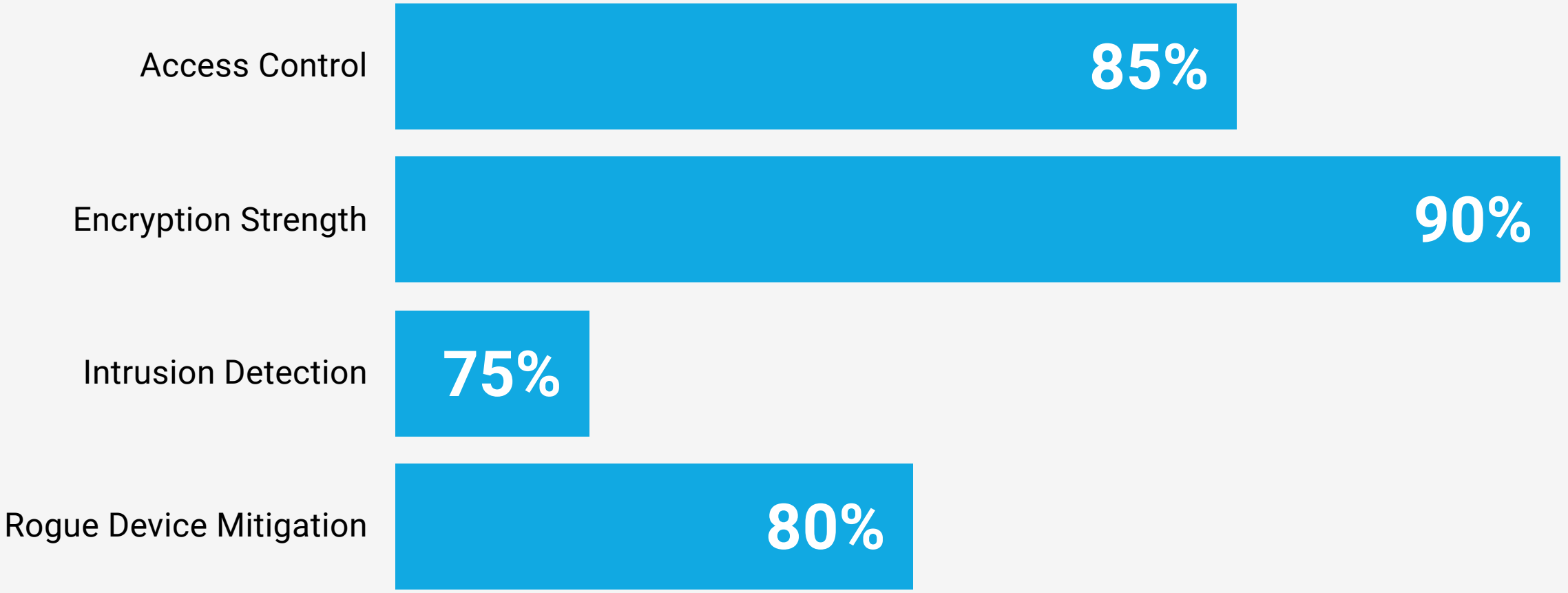
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Wireless Network Security



Wireless Network Architectures

Comparison of security factors between infrastructure mode and ad-hoc mode (0-100 scale)




EAVESDROPPING
MECHANISMS
PROTOCOLS
ISOLATION
AUTHENTICATION
IMPERSONATION
UNAUTHORIZED
WPA2
IPS
MIDDLE
NETWORK
DEFENSE PRACTICES
ACCESS
CONTROL
WPA3
ROGUE
THE
ATTACKS
ASSESS
WIRELESS
CONNECTIVITY
802
IDS
1X
INTERCEPTION
VULNERABILITIES
WEAK
IN
BEST
DEPTH
IEEE
MONITORING
SECURITY
11I
IN
WLAN
FIREWALL
ENCRYPTION
FI
AUDITING
MAN
RISKS
INTRUSION
LOGGING
WPA
COMPLIANCE
SPOOFING
WI
802
SEGMENTATION
POINTS
REGULATIONS
DETECTION
PREVENTION
AUTHORIZATION
STANDARDS
AUTHENTICATION

WPA, WPA2, and WPA3

Wireless Security Protocol	Key Security Features
WPA (Wi-Fi Protected Access)	Uses TKIP encryption, supports 802.1X authentication
WPA2 (Wi-Fi Protected Access 2)	Uses stronger AES encryption, supports 802.1X authentication

*IEEE 802.11i standard

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IEEE 802.11i and 802.1X



IEEE 802.11i

Defines security enhancements for wireless networks, including improved encryption and authentication mechanisms.



IEEE 802.1X

Provides port-based authentication for network access control, ensuring only authorized devices can connect to the network.

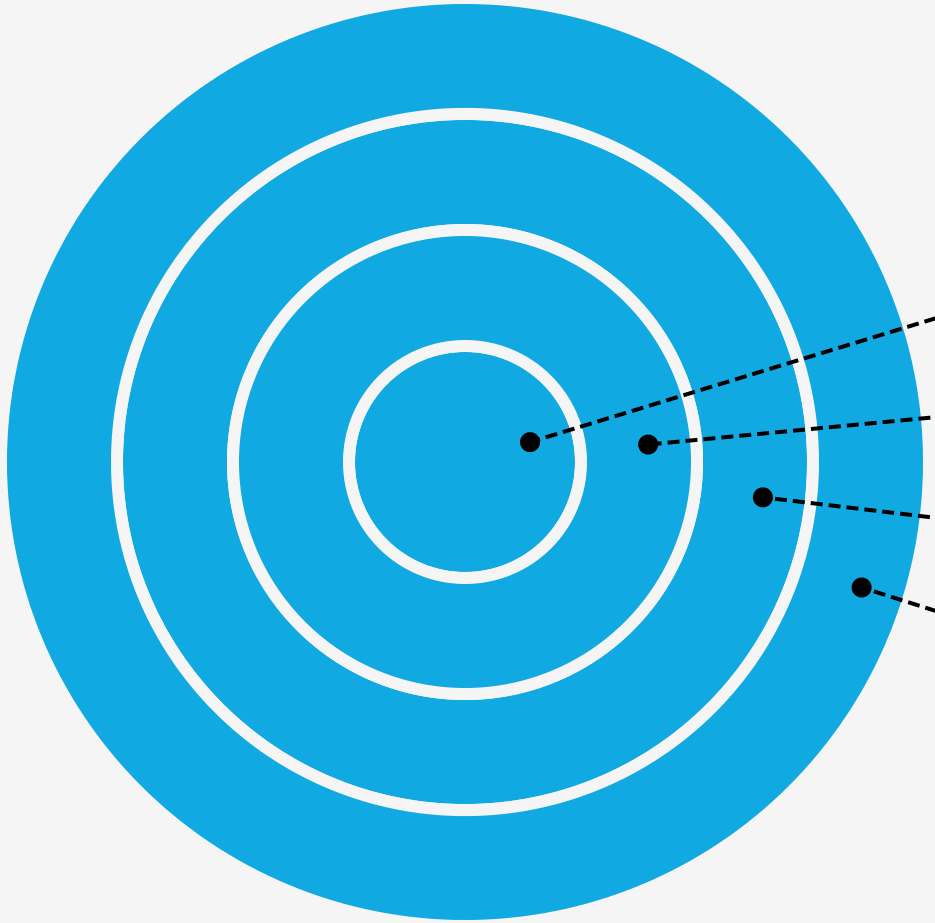


Secure Wireless Connectivity

The combination of these standards helps organizations achieve secure wireless communications by enforcing robust security measures.

IEEE 802.11i and 802.1X standards are essential for implementing secure wireless networks, protecting against unauthorized access and ensuring the integrity of wireless communications.

Zones of Control



Internal Zone

Securely houses mission-critical systems

DMZ Zone

Isolates public-facing services from internal network

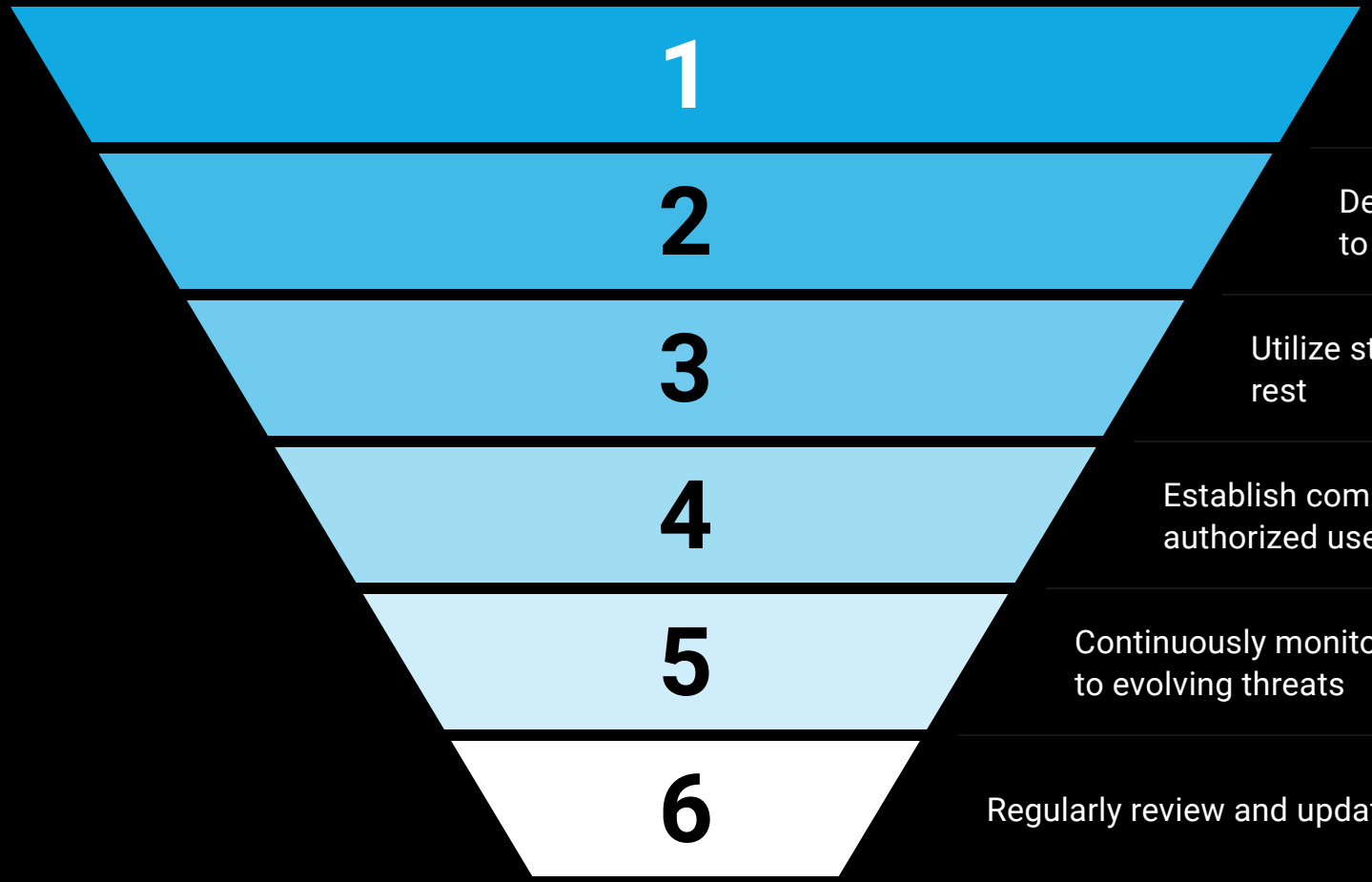
Guest Zone

Provides limited network access for visitors

Wireless Zone

Enforces strict security controls for wireless clients

Comprehensive Network Security



1. Implement robust firewalls to control network traffic and restrict unauthorized access
2. Deploy intrusion detection and prevention systems (IDS/IPS) to monitor for and mitigate cyber threats in real-time
3. Utilize strong encryption protocols to secure data in transit and at rest
4. Establish comprehensive access control policies to ensure only authorized users and devices can connect to the network
5. Continuously monitor network activity and security logs to detect and respond to evolving threats
6. Regularly review and update security measures to address emerging vulnerabilities

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